

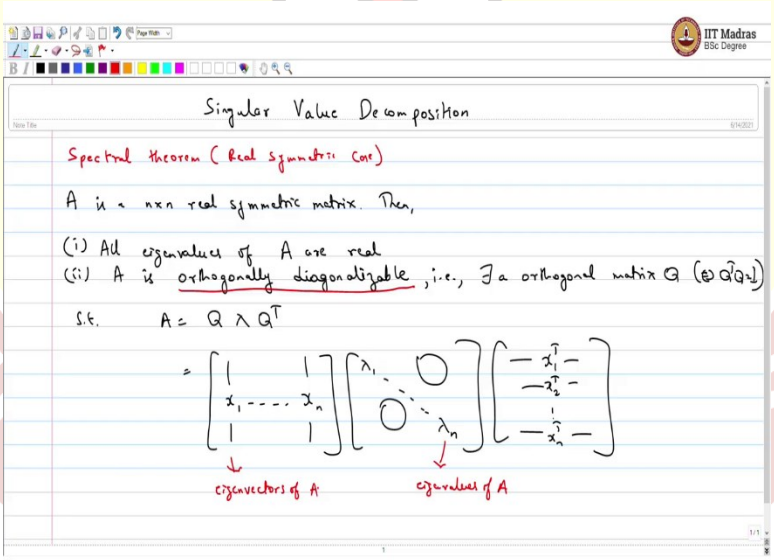
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor Prashanth LA
Department of Computer Science and Engineering
Indian Institute of Technology, Madras
Singular Value Decomposition

Welcome to lecture 6 in week 5 of machine learning foundations. In this lecture we will move on to a topic that is a logical continuation of what we have learned so far which is we will use spectral theorem. We saw spectral theorem in the previous lecture, which is, which states that any real symmetric matrix is orthogonally diagonalizable, but here we look at a more general class of matrices which are not necessarily one square, it could be a rectangular matrix and we will show that every rectangular matrix m rows and n columns can be decomposed using two orthogonal matrices and diagonal matrix and this is the so called SVD form which is the singular, which is short form for singular value decomposition. We will show this decomposition holds for any rectangular matrix. The proof along the way will use the spectral theorem at some point.

(Refer Slide Time: 01:11)

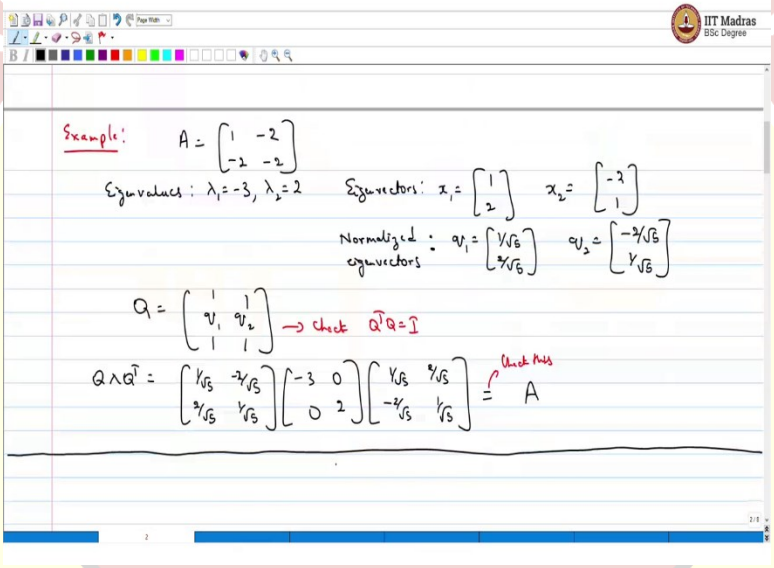


Before getting into singular value decomposition, I will do a quick recap of spectral theorem which is covered in a previous lecture video, so spectral theorem in particular for a real symmetric case. So, we have seen the more general version of this, but let us just recap this. Suppose A is a n × n real symmetric matrix, then A is orthogonally, let me remove this, then the first claim is all

eigenvalues of A are real, moreover, A is orthogonally diagonalizable that is there exists a orthogonal matrix. This is not a standard term.

So, Q this is equivalent to saying Q transpose Q = identity such that A can be written as Q λ Q transpose or let us put it more like this. So, then you have a diagonal matrix into the transpose. So, this is x1 transpose, x2 transpose and so on until xn transpose. So, here these are the eigenvalues of A and these xi are the eigenvectors of A. So, that is a spectral theorem of you.

(Refer Slide Time: 04:06)



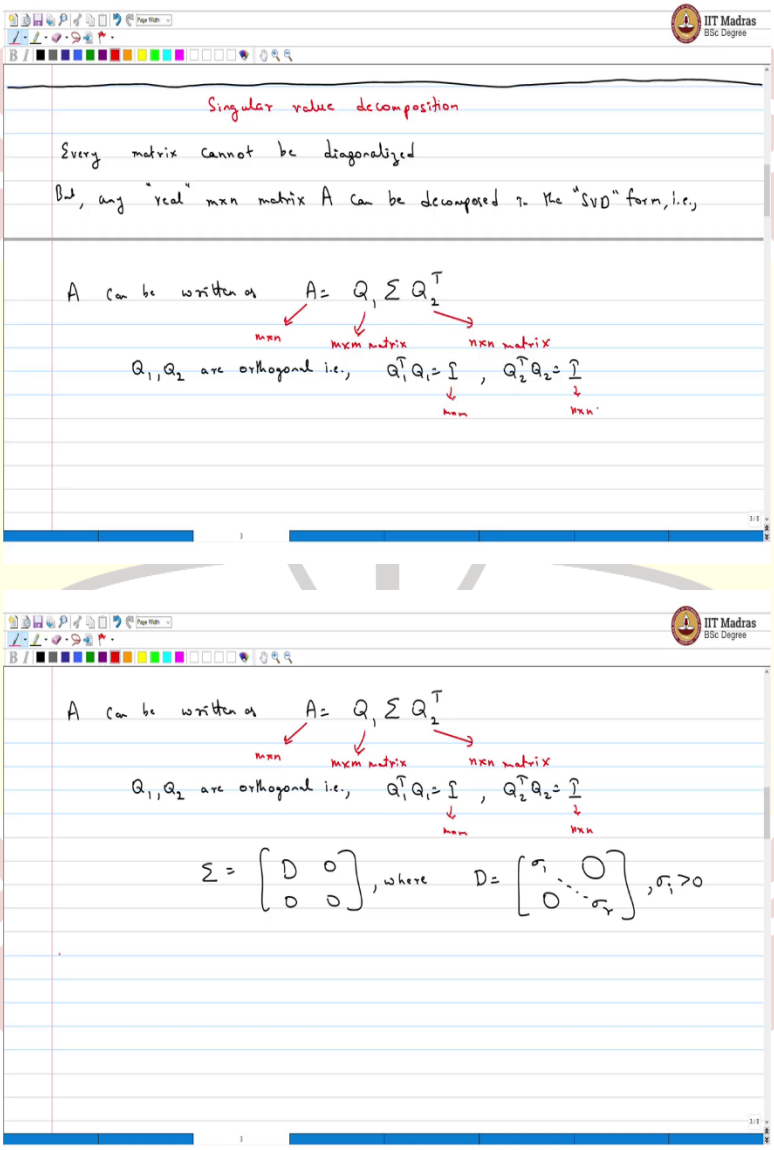
Just to reinforce the same idea, we will just do an example where I would not do all the calculations. This is fairly straightforward, just a 2 × 2 matrix, 1, - 2, - 2, - 2. So, what are the eigenvalues you can find out? I would recommend pausing the lecture video on working out this Q matrix and the eigenvalues on your own. So, the eigenvalues turned out to be - 3 and 2, λ 1 is - 3, λ 2 = 2. The corresponding eigenvectors are x1 is 1, 2, x2 is - 2, 1.

Now, normalized versions, just two eigenvectors, so it is, you do not have to take the pain of doing a Gram–Schmidt orthonormalization, just normalizing this, dividing it by the length we will do. So, we have 1 by √5 and 2 by √5 and q2 = - 2 by √5 and 1 by √5. Once you have this, how do you make up the Q matrix, take this q1, q2. It would be interesting to check Q transpose Q is identity.

So, when you are working out, calculating such things, sanity check is to do this Q transpose Q and ensure it turns out to be the identity. If it does not turns out, if it does not turn out to the identity

and that is something that has gone wrong in your calculation. You may have to revisit that. In this case, you can check this and now you can also check Q λ Q transpose is 1 by $\sqrt{5}$, 2 by $\sqrt{5}$, - 2 by $\sqrt{5}$, 1 by $\sqrt{5}$ into the eigenvalues - 3, this turns out to be, this product turns out to be, again, check this. So, so far, what we have seen is, this is the story for real symmetric matrices.

(Refer Slide Time: 07:23)



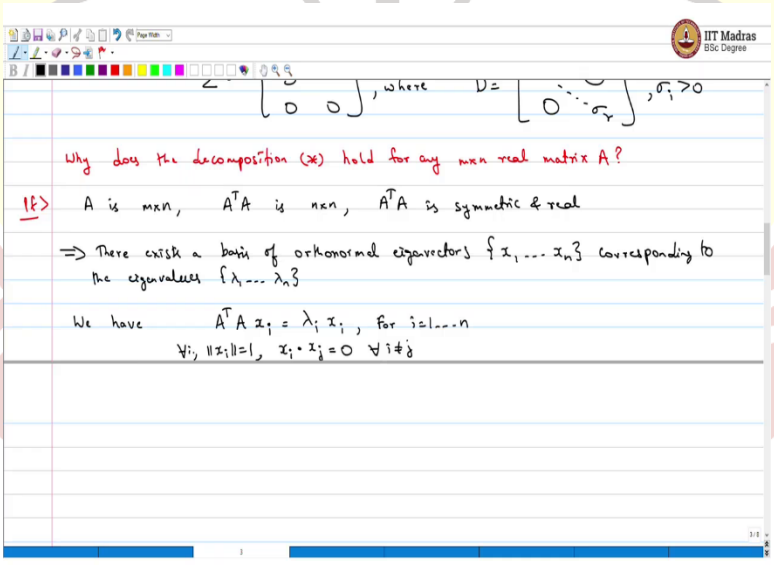
Now, we will see a more general decomposition. So, this spectral theorem or eigen decomposition is only possible for real symmetric matrices. It is not only possible. There are other matrices which satisfy, but a real symmetric matrix is guaranteed to have such a decomposition. Now, the next topic we are going to look at is singular value decomposition which applies to any matrix even if

it is triangular. We focus on the real case here. So, every matrix cannot be diagonalized. Seen enough examples.

However, any, we put this in quotes, real m × n matrix, even if it is a rectangular A can be decomposed in the SVD form. So, what is the SVD form? That is A can be written as A =, so here you have Q1 σ Q2 transpose. Now, if you look at this, Q1 is a m × n matrix. So, we get the dimensions right. A is m × n, Q is m × m and Q2 is n × n. The other facts are in the respective dimensions Q1, Q2 are orthogonal. So, Q1 , Q2 are orthogonal that is Q1 transpose Q1 = identity Q2 is also = identity, but this is identity, this is again m × m, but this is a different identity matrix. This is n × n.

And the last thing is matrix σ is a particular form where D is a diagonal matrix with positive entries σ r, where σ i is greater than 0. So, before delving deeper into this decomposition let us try to show that, I mean, given any real matrix, I mean, why does this decomposition even hold. I will try to argue that any real matrix can be broken up into three matrix product like this or decomposed into three matrices like this.

(Refer Slide Time: 10:47)



So, why does the decomposition star hold for any m × n matrix, so here is the proof. So, here is where we will invoke the spectral theorem at some point. But I would say just think about this decomposition again understand that this is a much more general decomposition than the one given by spectral theorem simply because we are not imposing symmetry here. A could be any matrix.

And moreover, we are not even requiring, we do not even require A to be a square matrix, any m × n matrix will do.

So, let us understand why this decomposition can be done for any matrix, a formal proof of that. So, what is A? A is m × n, so A transpose A is n × n and what kind of a matrix is A transpose A. For any matrix it does not matter what the matrix A is. For any given matrix A transpose A is symmetric. By definition A is having only real entry. So, when you do A transpose A you will still remain with real entries. You would not complex values.

So, given this fact implies there exists a basis of orthonormal eigenvectors, let us say x1 up till xn corresponding to, I mean, A transpose A can be diagonalized. So, that just means that I get enough eigenvectors and the number of eigenvectors A transpose A is n × n, so I get n eigenvectors independent one and I can also make them orthonormal. So, we have an orthonormal basis for Rn corresponding to the eigenvalues, let us say.

Now, here is where we will start analyzing. So, we have, so what do we have, A transpose A is the matrix that we diagonalized, xi is its eigenvector. So, A transpose A, I mean, when you multiply A transpose A with xi then you will recover λ i × xi for i = 1 until n. Moreover, these xis are orthonormal. So, length of xi is 1. And xi dot xj in this case it is just is 0 for all i ≠ j. For all i, I got, so this is an orthonormal basis. So, each eigenvector is of length 1 and they are orthogonal to each other.

(Refer Slide Time: 15:07)



So, continuing further along this. So, what we have is if I take A transpose A xi and dot product with xi this is λ i xi dot product of xi and that is just λ i since. I can compute this dot product in a different way also. A transpose A xi dot product with xi is A transpose A xi transpose xi which is a same as xi transpose A transpose A xi is same as A xi transpose A xi which is the same as finally and this is a non-negative quantity.

Just by definition that is the length of the vector A xi. So, it is non-negative. So, what we have is using what we call this equation 1, equation 2, 1 and 2 imply, so this dot product = λ i and this dot product is also non-negative. So, λ i has to be non-negative. Now, if you understand, we said that

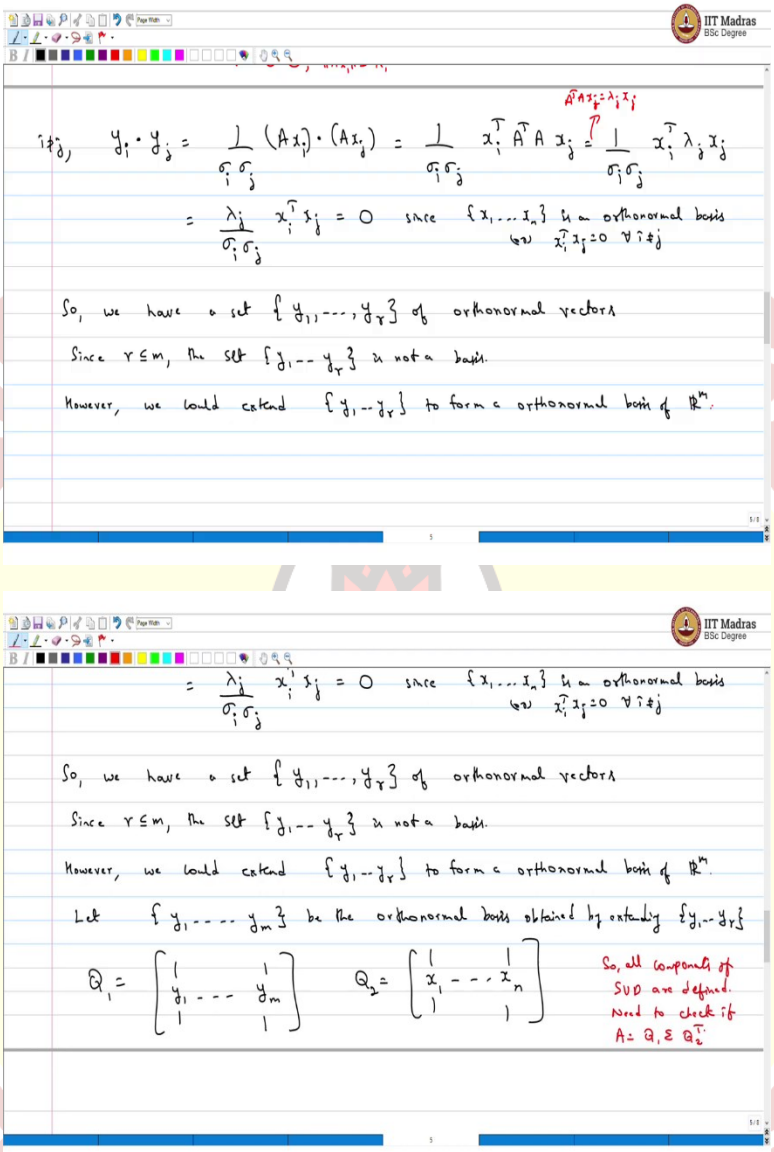
when we do the SVD you get a diagonal matrix with some numbers σ is, some r of them. So, where are these coming from and that is where we are going at.

So, far we have got, so what we have is let us order the λs, so order the eigenvalues of A transpose A which is seeing that say λ 1 is the biggest, λ 2 so on until λ r and λ r + 1 until λ n. And what we have is λ 1 greater than 0, so on until λ r greater than 0 and all the remaining eigenvalues are equal to 0. Now, you make up the definition.

So, σ i's are called singular values and for each i = 1 to r σ i = $\sqrt{\lambda}$ i. So, this is for the first, the, you take the positive eigenvalues of A transpose A. We just say that there r of them. And if there are r of them, take λ 1 until λ r, take their √and denote them by λ i. There is a particular reason why we have to take the square root, because finally that goes into the diagonal matrix. So, we have defined many bits at least the diagonal matrix is now clear. So, the matrix σ is now clear. So, we know what goes into σ.

Now let us see what goes into q1 and q2. Before that we need to a little more analysis. So, let yi = 1 by σ i A xi for i = 1 to r. So, let us just understand some basics here, xi is a n dimensional vector, A is m × n matrix. So, yi belongs to Rm. So, it is a m dimensional vector. What is the length of yi? We know that σ is positive. So, 1 by σ comes outside. This is length of A xi. But this length of A xi is the same as λ i. So, it is $\sqrt{\lambda}$ i/ σ i, but σ i is by definition λ, so length of yis are 1. Each yi is of length 1. So, A xi by definition is λ xi and then you know the length. It is pretty clear. A transpose A xi is λ xi, but from the calculation here that we have seen, let us see. From 1 and 2 mod A xi square = λ i. So, if you just go up here, we just calculated that λ i turns out to be the same as A xi square. So, that is yi.

(Refer Slide Time: 21:48)



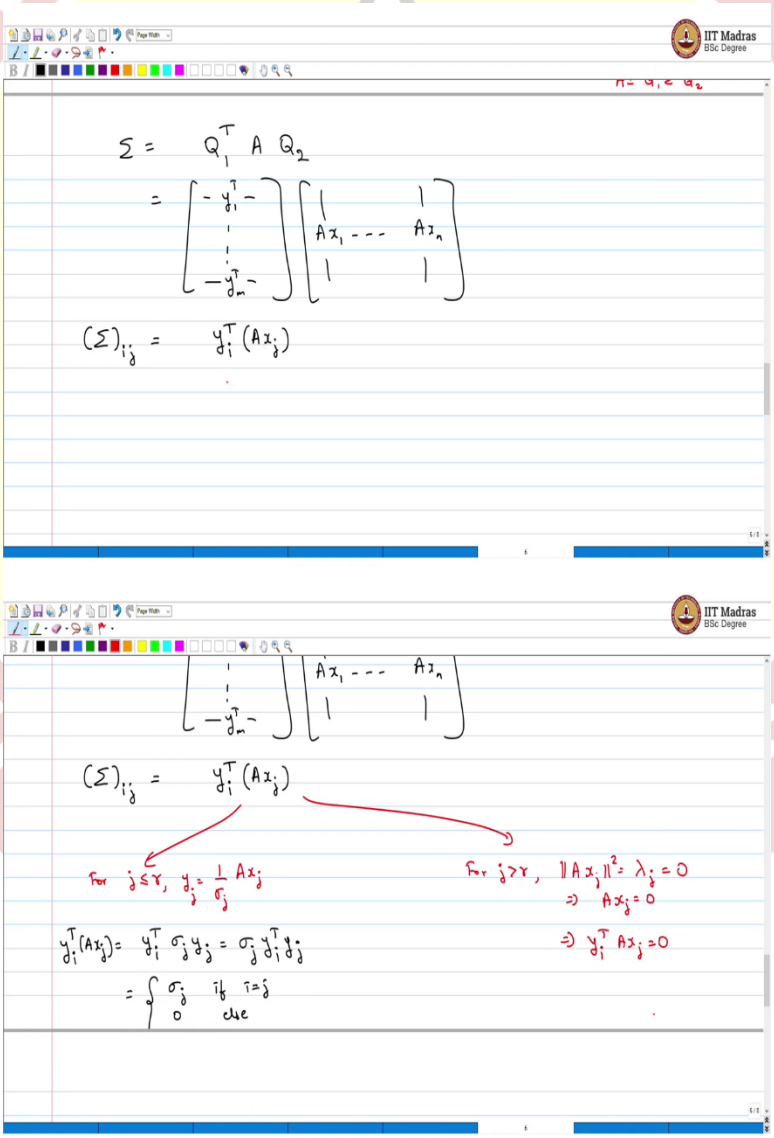
We will also show that if you take yi dot product with yj, i ≠ j, this is going to be 1 by σ i σ j A xi dot product with A xj. Now, this is the same as 1 by σ i σ j this = to dot product xi transpose A transpose A xj. But A transpose A xj, xj is a eigenvalue of, eigenvector of A transpose A. So, this = 1 by σ i σ j, xi transpose λ j xj simply because A transpose A xj = λ j xj.

So, this whole thing = λ j/ σ i σ j into xi transpose xj, but xi transpose xj is 0 since x1 to xn is an orthonormal basis or xi transpose xj is 0 for i ≠ j. So, what we have is, so we have a set y1 until yr is a set of orthonormal vector. This is not a basis, because yis are in Rm and we may have r may be less than m. Since r is less than or equal to m the set y1 to ym, yr is not a basis. However, we

could extend y1 to yr to form a orthonormal basis of Rm. So, I need m vectors. So, I could add the remaining m - r vectors. So, let us denote that extended set y1 until ym be the orthonormal basis obtained by extending y1 to yr.

Now, I am ready to define Q1. So, Q1 is going to be, we want to take these vectors y1 until ym and in Q2 I am going to put the eigenvectors of A transpose A. So, all components of SVD are defined. Now, what we need to check, need to check if A is Q1 σ Q2 transpose. So, with this definition of Q1, Q2 can I recover A and σ, can I recover A by this product?

(Refer Slide Time: 26:49)

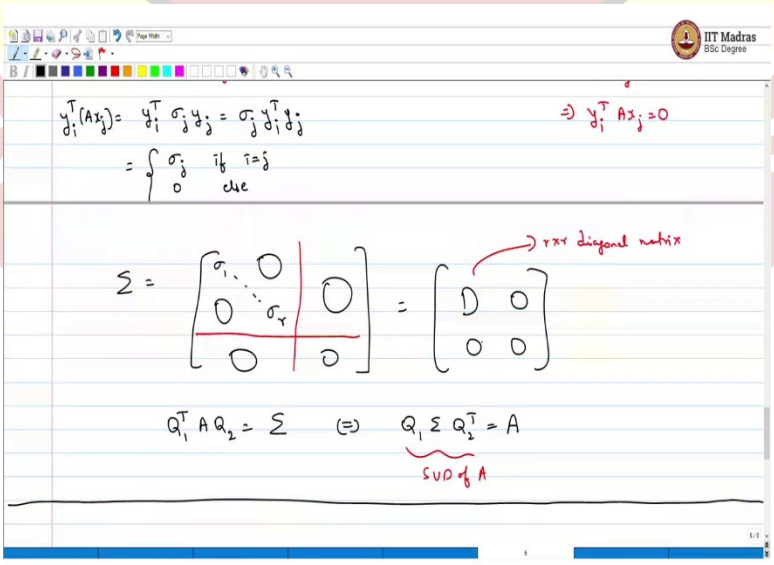


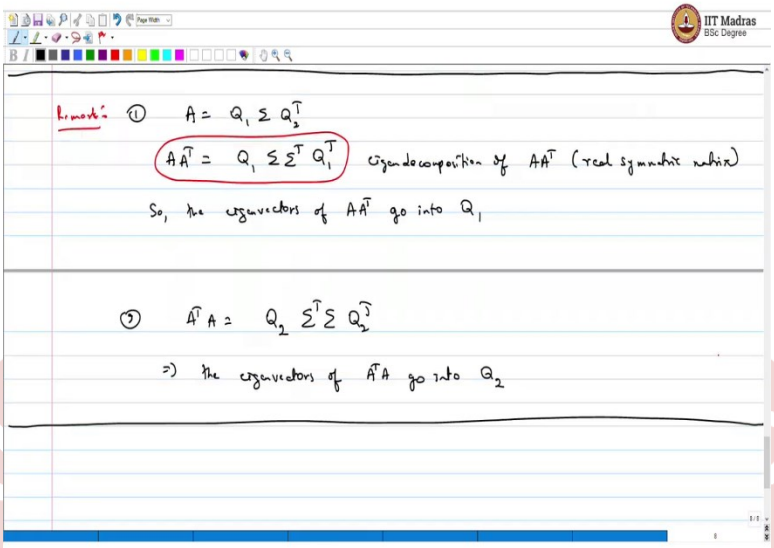
So, what is σ? σ is 1 transpose A Q2. Let us do this. Let us see if σ is the form that we want. So, what is Q1 transpose is just y1 transpose until ym transpose into A × Q2 is just Ax1 until Axn. Now, if I can show that σ has this form, this product Q1 transpose A Q2 has the form where it has a bunch of diagonal entries σ 1 until σ r and then 0s elsewhere then I am done. And that is the last bit of the proof remaining. So, this is the ijth entry. So, ijth entry of this product Q1 transpose A Q2 is I am going to take yi and multiply it with Axj. Now, let us understand this.

Now, there are two cases here. The very first case j less than equal to r. See the thing about yi says, if you recall until r, yi has a particular form. So, yi = 1 by σ Axi for all i less than or equal to r and beyond r, I got these vectors by extending to a full basis. So, let us calculate this for j less than equal to r. So, in this case, the j less than equal to r, yj = 1 by σ j into Axj. So, what happens to this product yi transpose A xj = yi transpose, so A xj is nothing but σ j yj. So, this = σ j yi transpose yj and this finally = the two things. It can be equal to σ j if i = j and 0 else. Now, the second case. So, I hope this was clear.

The case where j greater than r. Now, in this particular case what we have is j greater than r observed that what is A xj is nothing but λ j, but λ j for all j greater than R is just 0. You have λ r + 1 until λ n 0 by definition, which just means A xj = 0 which finally implies yi transpose A xj also equal to 0. So, that covers the entire deal.

(Refer Slide Time: 31:03)





So, what we have is, so the final thing is, so the form we get for σ is we have σ 1 until σ r then you have 0 elsewhere. So, this is the diagonal matrix that I was talking about. So, this is a r × r diagonal matrix. So, what we have shown is Q1 transpose A Q2 turns out to be the symmetric σ which is equivalent to saying you just pre-multiply with Q1 and post multiply by Q2 transpose and what you will get is Q1 σ Q2 transpose = A and this is SVD of A.

So, we have shown that any m × n matrix can be decomposed into this particular form, where Q1 transpose σ into product, I mean, into product of two orthogonal matrices with a diagonal matrix thrown in but these two orthogonal matrices live in two different spaces. One of them m × n, the other is n × n matrix. And also, we have seen that Q2 contains, by definition, when we saw Q2, what did Q2 contain, Q2 contained the eigenvectors of A transpose A.

Now, what does Q1 contain? That is a question we will answer next. I would say just think about this decomposition. And just make a remark about what goes into Q1 and Q2. Just from the decomposition if you have not seen this proof, we can infer what goes into Q1 and Q2. So, A is Q1 σ Q2 transpose. For A into A transpose Q1 σ σ transpose Q1 transpose.

And what you got is this is Eigen decomposition of A into A transpose. So, what goes into, so the eigenvectors of A into A transpose go into Q1, A transpose which is a real, just to emphasize the fact that it is a real symmetric matrix. So, A into A transpose is a real symmetric matrix. I can make its eigen decomposition using spectral theorem and σ into σ transpose is going to be a diagonal matrix. It is needless to say. So, σ into σ transpose is a diagonal matrix and Q1 is the

matrix where all the eigenvectors form. Along similar lines just from the decomposition, you can infer what is A transpose A. This turns out to be Q2 σ transpose σ into Q2 transpose implies the eigenvectors of A transpose A go in to Q2. So, those are the remarks. We will do an example now.

As summary of today's lecture before we wrap up, what did we learn today, first, we looked at a simple example where we worked out the Eigen decomposition of a real symmetric matrix, which is an illustration of the spectral theorem in action. And next we moved on to a very important topic which is the singular value decomposition or SVD in short. We showed that any rectangular matrix and say m × n matrix can be decomposed or written as a product of three matrices two of them orthogonal that is diagonal and these two orthogonal matrices are of different dimensions and this diagonal matrix is obviously a rectangular matrix.

And what goes into the diagonal matrix are what the so called singular values associated with the original m × n matrix. Now, the singular value decomposition holds for any increase any m × n matrix. We do not impose any additional structure, unlike spectral theorem, which required it to be, which required a matrix to be square, real, symmetric, all of that. We do not need any structure. And in the next lecture, we will look at a simple example which illustrates how to compute the SVD of a given matrix, given rectangular matrix. We will try to understand this by example in the next lecture.